

Boundary–Interior Closure, Neutral Empiricism, and α -Divergence Stances

A Programmatic Overview for Information Geometry & Dynamical Systems

Abstract

We present a boundary-first empirical program for dynamical inference. The primitive data are edge-level empirical objects (one-step marginals, and when available a multi-scale family across time steps). Interior dynamical structure—Markov chains, Markov semigroups, infinitesimal generators, and spectral elements—is treated as a *completion* (a regularity closure) that is earned under

0.1 Worked example: local linear least squares with residual-derived noise

This example exhibits the program in its most concrete form: a boundary-level empirical object (local stepwise conditionals), a completion step (constructing a Markov kernel and testing closure), and a continuous-time limit (a generator and its semigroup). The technical counterpart is Example 1 and Lemma 4 in the Elements document.

Given a time series $(x_k)_{k=0}^N \subset \mathcal{R}^d$ sampled at step Δt , fix a query point x and local weights $w_k(x) \geq 0$. Fit a locally affine predictor by weighted least squares,

$$x_{k+1} \approx A(x) x_k + b(x),$$

and define residuals $r_k = x_{k+1} - (A(x_k)x_k + b(x_k))$. A residual-derived noise model (e.g. Gaussian) estimates a local covariance $\Sigma_{\Delta t}(x)$ from the weighted residual second moment. This yields an explicit data-driven Markov kernel

$$K_{\Delta t}(x, dy) = \mathcal{N}(y; \mu_{\Delta t}(x), \Sigma_{\Delta t}(x)) dy, \quad \mu_{\Delta t}(x) := A(x)x + b(x),$$

and hence a Markov operator $(T_{\Delta t}f)(x) = \int f(y) K_{\Delta t}(x, dy)$.

Completion as closure. Rather than assuming a Markov semigroup, we treat Markov structure as a closure property to be *tested*: does the family $\{T_{\Delta t}\}$ approximately satisfy Chapman–Kolmogorov composition at small steps, and does it possess a stable generator limit as $\Delta t \rightarrow 0$? Divergences (including α -divergences) enter here as stance-dependent ways of measuring the deviation of empirical multi-step conditionals from the composed kernel.

0.2 Semigroup taxonomy from drift/noise scaling

In the LWLS construction the “destination” semigroup is determined by the limiting generator, which is in turn controlled by the *scaling of drift and residual noise with Δt* . Different modeling choices (and different empirical scaling regimes) therefore lead to different familiar semigroup classes, summarized in Table 1 (cf. Lemma 4).

Takeaway. The framework does not privilege a single interior dynamics class. It explains which semigroup class is warranted by the *boundary* statistics (including scaling diagnostics), and it makes non-uniqueness explicit when the boundary does not identify the interior uniquely. explicit consistency and limiting conditions, rather than assumed a priori. This viewpoint offers a systematic synthesis between reductionist and complexity approaches: reductionist structure is recovered when completion succeeds; irreducible pluralism is acknowledged, but disciplined, when completion fails or is non-unique. We also interpret Amari’s $\alpha \in [-1, 1]$ family of divergences as a parametric family of *empirical stances* governing which discrepancies are treated as epistemically costly, and we propose practical diagnostics for selecting α .

1 The boundary–interior principle

Many dynamical modeling pipelines implicitly begin with an *interior* object—a latent state space, equations of motion, or a Markov semigroup—and regard observations as a projection. In contrast, the boundary-first program takes as primitive what is empirically stable and directly comparable across representations: *edge-level* statistical information.

Boundary data. Let Y be an observation space and let $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ denote an empirical stepwise marginal at scale Δt . When multiple scales are available we consider an empirical family

$$\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in (0, \Delta t_0]} \subset \mathcal{P}(Y \times Y),$$

along with a reference/source law $\nu_Y \in \mathcal{P}(Y)$ and an information divergence D_f (in particular, Amari’s α -divergence family).

Interior structure as closure. An interior dynamical object (e.g. a continuous-time Markov semigroup) is not posited as ontology. Instead, it is treated as a *regularity closure problem*: when do the boundary data across scales admit a coherent interior completion, and when do they not?

The “necessary emptiness.” The gap between boundary data and an interior dynamics is not a deficiency of the framework; it is the correct epistemic state in complex systems. Completion is possible only under additional, empirically testable regularity. When completion fails or is non-unique, underdetermination is interpreted as an empirically meaningful feature (not an error), and the framework seeks invariants and partial comparability rather than forced uniqueness.

2 Cathedral summary: one object, three layers

The program is organized around a single core empirical object and three derived layers.

Core object (empirical, primitive)

$$(Y, \nu_Y, \{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in (0, \Delta t_0]}, D_f). \tag{1}$$

Layer A: neutral structure (always defined)

From a fixed $\Gamma_Y^{(\Delta t)}$ one obtains canonical one-step structure (e.g. by disintegration) and associated induced operators on a chosen space of observables. The *neutral framework* then organizes all realizations/factorizations compatible with the same boundary data into equivalence classes (“leaves”), and identifies invariants that do not depend on a privileged latent ontology.

Layer B: completion (conditional, earned)

- **Discrete-time:** A single Γ_Y always yields a forward Markov extension (one-step kernel), but a two-sided stationary completion requires an explicit balance/invariance condition on marginals.
- **Continuous-time:** A Markov semigroup arises only as a limit completion of the family $\{\Gamma_Y^{(\Delta t)}\}$ under semigroup-consistency and generator convergence conditions.

Layer C: spectral and geometric consequences (post-completion)

Generator spectra, resolvents, eigen-elements, and long-time asymptotics are defined only after a completion exists. Representation comparisons of spectral objects require (i) a neutral identification mechanism and (ii) a shared completed dynamics.

3 Edge germs, completions, and neutral invariants

The boundary data may be treated as a single mathematical object: a *germ* of edge statistics at small time scales. This packaging makes explicit (i) when an interior dynamics is determined, (ii) when it is underdetermined, and (iii) which quantities remain meaningful in the absence of a preferred completion.

3.1 The edge germ

Let Y be a standard Borel space. Fix a reference/source law $\nu \in \mathcal{P}(Y)$ and a family of edge marginals indexed by time scale,

$$\Gamma^{(\Delta t)} \in \mathcal{P}(Y \times Y), \quad \Delta t \in (0, \Delta t_0],$$

each with first marginal ν . The associated *edge germ* is

$$\mathcal{G}_Y := (Y, \nu, \{\Gamma^{(\Delta t)}\}_{\Delta t \downarrow 0}). \quad (2)$$

When only a single scale is available one may replace the germ by (Y, ν, Γ) , but continuous-time claims then require additional information.

3.2 Completions as a (possibly empty) family

An *interior dynamics on Y* is a family of Markov kernels $\{P_t\}_{t \geq 0}$ with $P_0 = \text{Id}$ and $P_t \mathbf{1} = \mathbf{1}$ (and, when needed, strong continuity on a chosen observable space X). Such an interior induces boundary edge marginals via

$$\Gamma_{\{P_t\}}^{(\Delta t)}(dy, dy') := \nu(dy) P_{\Delta t}(y, dy'). \quad (3)$$

We write $\partial(\{P_t\}) := (Y, \nu, \{\Gamma_{\{P_t\}}^{(\Delta t)}\})$ for the induced edge germ.

Definition 1 (Markov completion of an edge germ). *A (continuous-time) Markov completion of $\mathcal{G}_Y$ is an interior dynamics $\{P_t\}_{t \geq 0}$ such that $\partial(\{P_t\}) = \mathcal{G}_Y$, i.e. (3) matches the empirical $\Gamma^{(\Delta t)}$ for all Δt in the germ. The set of completions is denoted*

$$\text{Comp}(\mathcal{G}_Y) := \{\{P_t\}_{t \geq 0} : \partial(\{P_t\}) = \mathcal{G}_Y\}.$$

Necessary emptiness and underdetermination. In general $\text{Comp}(\mathcal{G}_Y)$ may be empty or may contain multiple non-equivalent completions. This expresses a precise form of “necessary emptiness”: boundary data do not automatically determine a unique interior dynamics. When a unique completion exists (up to standard equivalences), the framework recovers a reductionist-style interior description; when it does not, the appropriate empirical output is a structured description of the boundary object and its invariants.

3.3 Neutral invariants and representation-independence

Fix an information divergence D_f (including Amari’s α -divergences). The divergence specifies a geometry on spaces of boundary objects and thereby determines which distinctions are treated as empirically meaningful. We write

$$\text{Inv}_{D_f}(\mathcal{G}_Y)$$

for the collection of quantities or structures computable from $\mathcal{G}_Y$ that are invariant under admissible representation maps (e.g. coarse-grainings or sufficient statistics) that preserve the relevant D_f -geometry. Conceptually, these are the stable empirical contents of the program even in regimes where $\text{Comp}(\mathcal{G}_Y)$ is empty or non-unique.

3.4 Closure theorem schema (boundary $\rightarrow$ interior)

The completion problem becomes well-posed once the germ carries sufficient multi-scale regularity. A typical closure theorem has the following shape.

Closure schema. If the empirical family $\{\Gamma^{(\Delta t)}\}$ satisfies (i) an approximate Chapman–Kolmogorov (semigroup-consistency) property as $\Delta t \downarrow 0$, (ii) a small-time identity condition $\Gamma^{(\Delta t)} \Rightarrow \Delta_{\#}\nu$, and (iii) a generator-control condition (expressed on a chosen observable space X) yielding a well-defined dissipative limit, then $\text{Comp}(\mathcal{G}_Y) \neq \emptyset$ and there exists a C_0 Markov semigroup $\{P_t\}_{t \geq 0}$ whose skeleton matches the germ.

Conversely, any C_0 Markov semigroup preserving ν induces an edge germ satisfying these boundary regularity properties. In this sense, the existence of an interior semigroup is equivalent to boundary regularity plus an explicit closure condition, once the functional setting is fixed.

Table 1: How boundary-level drift/noise scaling selects the limiting Markov semigroup.

Boundary scaling signal	Limiting generator L (informal)	Familiar semigroup / forward equation
$\text{Var}(\xi_{\Delta t} \mid x) = o(\Delta t)$	$Lf = a \cdot \nabla f$	Transport/Liouville semigroup; $\partial_t \rho + \nabla \cdot (a\rho) = 0$
$\text{Var}(\xi_{\Delta t} \mid x) = B(x)\Delta t + o(\Delta t)$ (Lindeberg)	$Lf = a \cdot \nabla f + \frac{1}{2}\text{tr}(B\nabla^2 f)$	Diffusion / Fokker–Planck class; $\partial_t \rho = L^* \rho$ (heat, OU, Langevin as special cases)
<i>Heat:</i> $a = 0, B = 2\kappa I$	$Lf = \kappa \Delta f$	Heat semigroup; $\partial_t \rho = \kappa \Delta \rho$
<i>Ornstein–Uhlenbeck:</i> $a(x) = Ax, B$ const.	$Lf = (Ax) \cdot \nabla f + \frac{1}{2}\text{tr}(B\nabla^2 f)$	OU semigroup; linear Fokker–Planck
<i>Overdamped Langevin:</i> $a = -\nabla V, B = 2\beta^{-1}I$	$Lf = -\nabla V \cdot \nabla f + \beta^{-1}\Delta f$	Langevin diffusion; Gibbs stationary density (when normalizable)
Rare large increments; stable/jump scaling; Lévy limit	Integro-differential L with Lévy measure ν_x	Jump/Lévy semigroups; nonlocal forward equations (e.g. fractional Laplacian when symmetric stable)
CK composition fails systematically at observed level	No Markov closure on x alone (needs latent state or enrichment)	“No completion” on this boundary; or a projected/approximate Markov model

Recognition guide. In practice: (i) *approximately linear* drift $a(x) \approx Ax$ with *nearly state-independent* covariance suggests an Ornstein–Uhlenbeck (OU) limit; (ii) *near-zero* drift with *isotropic* covariance suggests a heat (pure diffusion) limit; and (iii) drift that is *approximately a negative gradient* $a(x) \approx -\nabla V(x)$ with $B \approx 2\beta^{-1}I$ suggests an overdamped Langevin limit. Heavy-tailed/rare large increments indicate jump (Lévy) limits, while systematic Chapman–Kolmogorov failures indicate that the observed variables do not close Markovianly (latent state or enrichment required).

4 What is resolved: a synthesis of reductionist and complexity approaches

The boundary-first closure program offers a resolution/synthesis in the following sense:

1. **Reductionist structure is recovered when warranted.** If the boundary family admits a stable completion (e.g. a C_0 Markov semigroup), one inherits the full reductionist toolkit: generators, spectra, invariant measures, and mechanistic interpretation.
2. **Complexity pluralism is retained when necessary, but disciplined.** When completion fails or is non-unique, the framework does not force a mechanism. Instead it supplies structured comparability through neutral invariants, divergence geometry, and equivalence classes of representations.
3. **Incommensurable model classes become comparable.** Mechanistic, data-driven, and

stochastic-closure models can be compared through a common currency: their induced boundary objects $\Gamma_Y^{(\Delta t)}$ across scales and the divergence geometry on the space of such objects.

5 Amari α -divergences as empirical stances

We adopt Amari’s convention $\alpha \in [-1, 1]$. Within this program, α is interpreted as a *stance parameter*: it encodes which direction of discrepancy is treated as epistemically costly, and therefore what kind of closure is preferred in moving from boundary to interior.

5.1 Three canonical stances

- $\alpha = 0$ (**structural realism / invariance-first**). A neutral, self-dual stance emphasizing representation-stable structure. One privileges invariants across admissible transformations and avoids directional bias between “world-to-model” and “model-to-world” mismatch.
- $\alpha \rightarrow +1$ (**realist-leaning / coverage-first**). A stance prioritizing coverage of observed transitions, including rare events. Epistemic cost is high for models that under-assign probability to observed behavior. This tends to favor conservative closures that do not rule out empirically observed edges.
- $\alpha \rightarrow -1$ (**pragmatic / parsimony-first**). A stance prioritizing sharpness, stability, and avoidance of unsupported transitions. Epistemic cost is high for models that posit transitions not supported by boundary data. This often yields cleaner coarse dynamics and more interpretable spectral structure, potentially at the cost of tail coverage.

5.2 Empirical diagnostics for choosing α

The stance interpretation is not merely philosophical: it yields testable, data-driven criteria. For a given α , assess:

1. **Completion stability across scales:** does the family $\{\Gamma_Y^{(\Delta t)}\}$ satisfy approximate semigroup consistency more strongly, and does a generator limit appear more stable?
2. **Spectral robustness:** are dominant spectral elements reproducible under resampling, subsampling, or coarse-graining?
3. **Rare-event sensitivity:** does α materially change inferred tail transitions or metastable structure? If so, is that variation stable and predictive, or noise-amplifying?
4. **Representation invariance:** are neutral equivalence classes (leaves) coherent and stable under admissible maps, or brittle?

6 Figure: α as a stance parameter in boundary–interior closure

Figure 1 summarizes the stance interpretation and its diagnostic consequences.

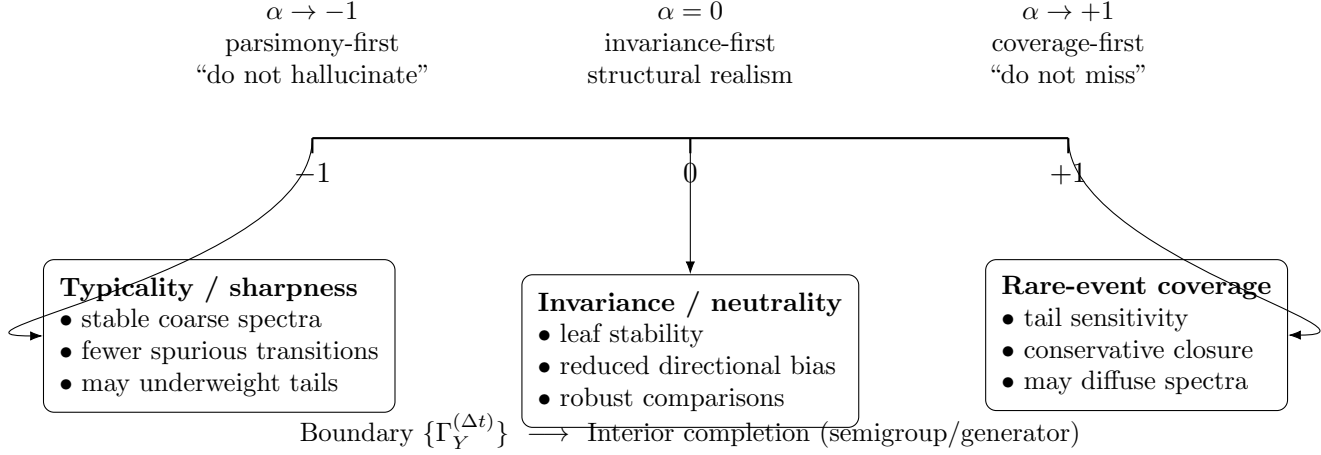


Figure 1: α as a stance parameter in the boundary–interior closure problem. Extremes $\alpha \rightarrow \pm 1$ express directional risk preferences (parsimony vs coverage), while $\alpha = 0$ emphasizes representation-stable structure.

7 Closing statement

The boundary-first program reframes long-standing tensions between reductionist and complexity approaches. Interior dynamics are not a precondition for inference, but a closure that may be earned from multi-scale boundary data. Divergence choice—especially within Amari’s α -family—is accordingly interpreted as an explicit empirical stance governing closure preferences, with empirical diagnostics available to adjudicate these stances in practice.

Pointers to the technical companion

This programmatic overview is intended to run in parallel with a technical “elements” document. In particular, the neutral objects and completion-as-closure statements are made precise in Definition 30 and the Markov completion sections (see §1.6 and §1.7), with core invariance and embedding results collected in §2.

Acknowledgment of scope. This document is intentionally programmatic: it sets the architectural logic and interpretive stance. Technical statements (existence, uniqueness, stability of completions; invariance theorems; spectral consequences) are to be supplied in the companion technical manuscript once hypotheses and function-space choices are fixed.